Zoo-511T

Principles Of Systematics

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SUBJECTIVE QUESTIONS:

Social variation of insects.

In social insects there are definite groups of individuals in a colony. • Each type of individual in a colony performs specific function. • For examples in a colony of Honey bee: • Queen, males (drones) and workers, soldiers

Measures of variability.

1. Mean deviation
2. Variance
3. Standard deviation
4. Standard error
5. Range

Lateral line system and swim bladder of pisces.

The lateral line system functions
as a sensory organ to sense the disturbances in the nearby environment. The swim bladder, also known as the gas bladder or air bladder, is a gas-filled organ found in most fish. It plays a crucial role in maintaining buoyancy, allowing fish to remain at their current water depth without having to expend energy in swimming

Classification of lady beetle

kingdom: animalia

phylum: arthropoda

class: insecta

order: coleoptera,

family :coccinellidae.

Morphology of spider telson.

The telson or “sting” is the final structure of the metasoma that comprises a bulbous vesicle, which contains the venom gland, and a needle-like aculeus to perforate and inject the venom. All scorpion species possess venom, which is used to kill or paralyze prey and as a defense against predators

Composition of silk.

Silkworm consists of two main proteins, sericin and fibroin, fibroin being the structural center of the silk, and serecin being the sticky material surrounding it Chemical composition: fibroin: 70-75%, silk gum: 25-30%, waxy substance 2-3%, natural color 1-1.5%, mineral matter 0.5-1

Standard deviation.

• It is positive square root of variance. • Standard deviation "provides a sort of average of the differences of all scores from the mean. • It is a measure of the dispersion of scores around the mean. • Generally, a low standard deviation means that a set of scores is not very widely dispersed around the mean. • While a high standard deviation indicates that the scores are more widely dispersed • Its advantage over variance is that its physical unit are same as that of individual observation

Water vascular system of echinoderms.

Echinoderms have a unique system for gas exchange, nutrient circulation, and locomotion called the water vascular system. The system consists of a central ring canal and radial canals extending along each arm. Water circulates through these structures allowing for gas, nutrient, and waste exchange.

main components are: madreporite, stone canal, ring canal, radial canal, lateral canal, tube feet, ampullae

Life cycle of lac insects.

After attaining the maturity, males emerge out from their cells and walk over the lac incrustations.
1.The male enters the female cell through anal tubular opening and inside female cell it fertilizes the female

2.Female lay eggs into incubating chamber inside her body. (June- July/oct-Nov)

3.After six weeks larvae are formed. (Feb-March)

4.These larvae then come out of female body and fix themselves to the Twing of plant (secrete resinous substance (LAC) around themselves and sit inside it and undergo molting) (3 times)

5.Adult

Skewed and normal data?

In normal data values of means, median mode are equal but this not the case for asymmetric or skewed data. • Normal data follow a bell shape curve. In skewed distribution one tail is larger than other. • In normal data most of the individual values are close to the mean of the data. However, in

skewed that the difference between individual values and mean values is higher. • Extreme values are almost negligible in normal data but higher in Skewed Data • Values of standard deviation are less for normal data compared to skewed data. • Measure of central tendency for normal data is mean but it is median for skewed data.

Prokaryotes and eukaryotes

Prokaryotes are unicellular organisms without a defined nucleus or membrane-bound organelles, such as bacteria (e.g., Escherichia coli). Eukaryotes have a well-defined nucleus and membrane-bound organelles, seen in organisms like plants (e.g., Arabidopsis thaliana) and animals (e.g., Homo sapiens). This structural difference leads to more complex cellular processes in eukaryotes compared to prokaryotes.

Aims of systematics:

Invent worlds kinds of organisms (flora and fauna) • Provide method for identification and communication • To produce coherent and universal system of classification and • Demonstrate evolutionary implications of biodiversity

Body shape and symmetry in echinoderm.

Symmetry usually radial, nearly always pentamerous. • Body shape globular, star-like, spherical, discoidal, or elongated

Test name of parametric test.

Parametric tests are used for Normal data • T-test • Z-test • F-test • ANOVA • Regression • Correlation

Crop and gizzard function in birds.

The crop stores and softens the food; however, the gizzard helps in crushing and churning the food

: From which plant part aphids can access phloem vessel?

Aphid feed from the phloem of plants which they trap into with the styelets of their proboscis. They gain access to the phloem vessels from 3 main parts of the plant, stems, leaves, and root

Characteristics of amphibians.

• They occur in fresh water and moist land. • Body is divisible into head and trunk.• The skin is smooth or rough having glands which keep it moist. •The gills are present at least in the larval stage • The respiratory organs are lungs • The heart is three chambered, having two auricles and one ventricle. • Kidneys are mesonephric. •Ear consists of internal and middle ear. The eyes have eyelids • Fertilization is external

Structure and morphology of scorpion chelicera.

The chelicera of scorpion is present in cephalothorax region. • They are scorpions jaws. • Each chelicera comprises three parts a coxa, tibia which has a fixed finger and the tarsus forming a moveable finger

Non-probability sampling and its types.

• Quota sampling

: In this type of sampling we divide the population intro different groups or strata based on some specific criterion but selection of individual from each group is not random •

Purposive sampling:

In this technique researcher selects sample based on their knowledge and professional judgment. Purposive sampling can also be known as judgmental, selective or subjective sampling. For example purchase of car from market based on expert opinion or selection of staff for biology teaching in a college by a panel of subject experts.

Drawback of the two-kingdom system.

The two-kingdom system fails to adequately classify microorganisms like bacteria and fungi. It struggles with organisms such as protists and algae that don't fit neatly into plants or animals. Additionally, it ignores genetic and evolutionary relationships among organisms.

Characteristics of Aristotle in taxonomy.

Aristotle also made distinction in insects • He first used the following terms: Coleoptera and Diptera • He established numerous collective categories, or genera using differentiating characters • Aristotle made further categories of animals • His thinking dominated for next two thousand years • After his death no system of classification were recognized .

Define sexual dimorphism.

> Sexual dimorphism refers to the differences in appearance between males and females of the same species, such as size, color, or morphology. Examples include the larger antlers of male deer and the colorful plumage of male peacocks compared to their female counterparts.

Define random sampling.

If every individual of target population has an equal chance of selection then it will be called random sampling. Simple random sampling is the easiest form of probability sampling. • No biasness and expert opinion is required in this method

Sympatric speciation.

A speciation in which a species is evolved from a single common ancestral species while inhabiting the same geographical range. • Share the same habitat but become reproductively isolated from each other.

Systematic sampling with two examples.

n this sampling method first sample from the population is taken randomly but after that samples are selected systematically at a regular interval. Examples: • Shopping centre END • Selection of plants • Selection of soil sample

Three domain system.

All organism belong to one of the three domains, depending upon their characteristics. Archea, Eubacteria , eukaryote. Introduced by Charl Woese et al. In 1990

Importance of systematic classification

Systematic classification helps us organize and understand the diversity of living things by grouping them based on their relationships. It makes it easier to identify, name, and study different species. This system also helps us predict traits of organisms based on their group.

Explain the role of pheromones in insect behavior.

Pheromones have been used to attract spined soldier bugs to the garden.
Most aphids have conicles (excrete pheromones.).

Describe the process of metamorphosis in insects.

Metamorphosis in insects is the process where they transform from larvae to adults. First, an egg hatches into a larva, which then grows and molts several times. The larva becomes a pupa, undergoing significant changes inside a cocoon or chrysalis. Finally, the pupa emerges as an adult insect, ready for reproduction.

What is the significance of the cuticle in arthropods?

The cuticle in arthropods is a tough, protective outer layer that provides structural support and shields against environmental damage. It helps prevent water loss and protects against predators and physical injury. Additionally, it aids in the attachment of muscles, enabling movement.

Differentiate between holometabolous and hemimetabolous development.

Holometabolous development involves complete metamorphosis with distinct life stages: egg, larva, pupa, and adult, as seen in butterflies. Hemimetabolous development features incomplete metamorphosis with gradual changes: egg, nymph (or naiad), and adult, as seen in grasshoppers.

Explain the function of malpighian tubules in insects.

Malpighian tubules in insects function as excretory organs that remove waste products from the blood and convert them into a solid or semi-solid form for excretion. They help regulate water balance and eliminate nitrogenous waste efficiently.

Describe the feeding habits of predaceous beetles.

Predaceous beetles are carnivorous and feed on other insects and small invertebrates, helping control pest populations. They use their strong mandibles to capture and consume their prey.

What is Batesian mimicry?

Batesian mimicry is a type of mimicry in which platable animal mimics the bad-tasting or noxious prey so that predator may not distinguish b/e them

Define diapause and its significance in insect life cycles.

Diapause is a period of dormancy in insects that helps them survive unfavorable environmental conditions, such as extreme temperatures or lack of food. It allows insects to pause their development and resume activity when conditions improve, ensuring their survival and reproductive success.

Explain the role of juvenile hormone in insect development.

Juvenile hormone (JH) regulates insect development by maintaining larval characteristics and preventing premature metamorphosis. It ensures that larvae continue to grow and molt without transforming into pupae. As JH levels decrease, insects transition to the pupal stage and then to adulthood.

LONG QUESTIONS;

Glands in mammals.

These animals are warm blooded, hairy and have mammary or milk producing glands, (mammary glands). • They are homothermous. • Oil glands (sebaceous glands) and sweat glands (sudoriferous glands) are present in the skin

Characteristics of molluscs.

Phylum Mollusca • They are mostly marine. • They have segmented body with distinct head, muscular foot and visceral hump. • They show bilateral symmetry. • The shell may be external, internal or absent. It is secreted by mantle and made up of calcium carbonate

Describing new species.

To describe a new species, you need to have the appropriate credentials, experience, and knowledge to do so. Carrying out the necessary measurements (scales, size, morphological characteristics, etc.) needed in a description

Contribution of Aristotle.

He established numerous collective categories or genera using differentiating characteristics Basic Terminologies • Blooded and bloodless • Two-footed vs four footed • Hairy vs feathered • With or without an outer shell • Note: He failed to provide orderly and fully consistent classification of animals

Characteristics of reptiles.

Reptiles are vertebrates. They have backbones. • Their bodies are completely covered with scales. • They are cold-blooded. • Reptiles produce shelled eggs or bear live young. • All species fertilize eggs internally. All species of reptiles have at least one lung. • Reptiles are the creeping and burrowing cold blooded vertebrates bearing epidermal scales. •

Dichotomous key and its benefits.

A dichotomous key is a device for identifying unknown organisms to some taxonomic levels such as species, genus or family level. • These most commonly published keys were first published by Jean B.

Lamark 1778. • If we continue to select correct option of couplets for unknown organism, correct identification of animals will be achieved

Uses of formalin as preservatives.

It is used for vertebrate only • Not used for long term storage • Acidic and difficult to handle • Used when colour of animal is important • Alcohols dissolve most colour immediately • Penetrate more easily and internal organs remain in better condition

Classification of reptiles.

Kingdom: Animalia

Phylum: Chordata

Class: Reptilia

Orders:

- Crocodylia (Crocodiles, Alligators, and Caimans)

-

**Species: *Boa constrictor*

Cluster sampling.

It is a sampling technique where the entire population is divided into groups, or clusters and a random sample of these clusters are selected. • Members of one cluster have similarity with each other and also with other clusters; minor differences can be there between different clusters. • Individual selection from each cluster is random. • Cluster of plants for study

.8. Define ambulacral system and its function in locomotion.

Notes on ANOVA.

Analysis of variance (ANOVA) is a technique used to compare means of more than two samples. • This technique can be used only for numerical data as it is a parametric test."

Assumptions •

Number of samples must be more than two. • Data should be normally distributed. • Samples must be independent.

• Homogeneity of variances

. ANOVA Examples • A group of psychiatric patients are trying three different therapies: counseling, medication and biofeedback. You want to see if one therapy is better than the others. • Students from different colleges take the same exam. You want to see if one college outperforms.

Fish preservation.

After capture fishes are placed in 10% formalin for quick killing. • This is highly painful • Fishes dies with its parts well spread off and body straight and well stretched. Methods of Animal Preservation •

Examination and counting of rays, fins etc. very easy. • 30 cm fish, Fish in formalin for one week, in water for one day (leech out formalin), use alcohol for long term preservation.

Phenogram and its rules.

• A tree diagram depicting taxonomic relationships among organisms based on overall similarity of many characteristics without regard to evolutionary history.

Start from lowest value • Try to make maximum separate clusters • Do not repeat • Incase of same number multiple time, you can chose any combination • Value of previous similarity should be higher than afterwards.

Characteristics of Annelida (2)

Their body is bilaterally symmetrical • They have a long and segmented body. • • They are triploblastic. . The body is covered with a thin cuticle. • They are coelomate. • Annelids live in moist environments, moist soil, freshwater and marine water. • They have parapodia and chitinous setae, used for locomotion. • Their body appears red due to the presence of haemoglobin

Mating rituals in spider lynx

The male performs a dance before the female using it legs, abdomen and especially his palps. • The female spider spins a cocoon that she attaches to a plant. long spines on their • • She will guard the eggs until they hatch. I Their abdomen is like a pencil ending in a point. • Some oxyopids may produce several egg sacs, while others only one. • Lynx's are very protective of their egg sacs, guarding them

Topotypes and its rejection

Specimens collected at type locality are called Topotypes. • In the case of parasites, the term type host is used to indicate the host organism from which the type specimen was obtained. Damage holotype • Information about environment • Temperature/Humidity/soil type • Missing type locality • Article 72H contains advice on designations or restriction • Designations or restriction not done routinely but by specialists by their revisionary work

Post mortem variation

it is impossible to prevent postmortem changes in some animals. • Deep yellow plumage of twelve wire bird of paradise----white • Chinese jay plumage changes from green to blue in collection • Wasps over-expressed to cyanide turn bright red • Insects • Spiders

54. Simpson and Friedrich contribution in taxonomy

Drawbacks of Biological species concept

• Distinguishing between species on the basis of reproductive separation is problematic because reproductive isolation is often incomplete. Not applicable on apomictic species • Do not fulfill interbreeding criterion • Apomictic groups. Evolution is gradual and continuous process. However, fossil record is not complete

Feeding of jumping spider

Jumping spiders hunt and feed on small insects• All are carnivorous, but a few species also eat pollen and nectar

Assumptions about evolution of species

Assumptions about the evolution of species include that species evolve through natural selection acting on heritable traits, that genetic variations arise through mutations and recombination, and that evolutionary changes occur over long periods, shaping the diversity of life.

Applications of molecular markers

Fingerprinting • Diversity studies • Genetic maps • Gene tagging • F1 identification • Seed testing • Comparative maps

T test sample size

For two sample t -test, the two populations must be independent • It is not necessary that sample size should be same for both samples

Non genetic variations

Individual variations in Time • Social Variations • Ecological Variations • Traumatic variation Non-Genetic Variations • Post-morten variations

Ecological behavior: This refers to how organisms interact with their environment, including how they find food, mates, and shelter. It helps them adapt to their surroundings and survive.

Nematode habitat: Nematodes live in various environments, including soil, water, and as parasites in plants and animals. Their habitat varies based on their species and ecological role.

Importance of biodiversity conservation: Biodiversity conservation maintains ecosystem balance, supports life-sustaining processes, and provides resources and benefits to humans. Protecting diverse species helps ensure ecological stability and resilience.

Natural selection process: Natural selection is when organisms with traits better suited to their environment survive and reproduce more successfully. For example, peppered moths evolved darker coloration to blend in with polluted trees, improving their survival.

Adaptations of insects for terrestrial life:** Insects have developed adaptations like a tough exoskeleton to prevent water loss, specialized respiratory systems for breathing air, and legs for walking and jumping on land.

Economic importance of insects:** Insects contribute to agriculture by pollinating crops, breaking down organic matter, and controlling pests. They also provide food for other animals and are used in research and medicine.

Role of insects in pollination:** Insects like bees and butterflies transfer pollen between flowers, aiding in plant reproduction. This process is crucial for the production of many fruits, vegetables, and seeds.

Structure and function of insect antennae:** Insect antennae are sensory organs that detect chemicals, vibrations, and touch. They help insects find food, mates, and navigate their environment.

Impact of climate change on insect populations:** Climate change affects insect populations by altering their habitats, food sources, and breeding cycles. It can lead to shifts in distribution, population declines, or increases in pest species.

Role of hormones in insect metamorphosis:** Hormones regulate the stages of insect metamorphosis, controlling changes from larva to pupa and then to adult. They coordinate growth, development, and transition between life stages.

Significance of genetic diversity in populations:** Genetic diversity helps populations adapt to changing environments, resist diseases, and avoid extinction. It ensures that a range of traits is available for natural selection to act upon.

Structure and function of the insect exoskeleton:** The exoskeleton is a hard outer covering that provides protection, support, and prevents water loss. It also serves as a site for muscle attachment and helps in movement.

Types of insect pheromones and their roles:** Insect pheromones are chemicals used for communication, including attracting mates, marking territory, or signaling danger. Examples include sex pheromones for attracting mates and alarm pheromones for warning of threats.

Importance of integrated pest management (IPM):** IPM combines various pest control methods, such as biological, chemical, and cultural practices, to manage pest populations sustainably and minimize environmental impact.

. Methods used for studying insect behavior:** Methods include observation in natural settings, controlled laboratory experiments, and using tools like video recordings and tracking devices to analyze and understand insect behavior.

Role of insects in nutrient cycling:** Insects break down organic matter, such as dead plants and animals, into nutrients that enrich the soil. This process helps recycle nutrients and supports ecosystem health.

Density dependent variations and allometric variations:** Allometry results in misappropriate size of some structure in relation to rest of the body. • This results in animals of different sizes----Allometric variability. When newly hatched nymphs of locust are reared under crowded conditions----Gregarious phase • Less crowded condition-- Transitional phase • Separate rearing---Solitary phase • These phase differ in anatomy, colour and behavioral characteristics---- distinct species

Characteristics of high weight characters:** Complexity • Constancy • Consistency • Not affected by ecological shift • Apomorphy • Not a specific adaptation • e.g. Morphological and behavioral characters. High weight characters are traits with a significant impact on an organism's fitness or survival, such as large size or strong physical attributes, which often have a major influence on reproductive success.

Contributions of applied biology:** Applied biology uses biological principles to solve practical problems in fields like agriculture, medicine, and environmental management. It helps improve crop yields, develop medical treatments, and address ecological issues.

Name different tests:**

independent t -test/two sample test • Paired t-test

Independent t-test

• For two sample t -test, the two populations must be independent • It is not necessary that sample size should be same for both samples. • Examples: Test the hypothesis that means heights (cm) of plants randomly selected from two wheat fields are same.

Molecular and biological characters:** Molecular characters involve genetic information, such as DNA sequences, while biological characters refer to physical traits and behaviors of organisms. Both are used to classify and understand species.

"Believe in yourself, work hard, and success will follow."

BS-ZOOLOGY 5th Semester

